

Hands-on introduction to (one) approach for evaluating informative hypotheses

Information criteria

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Example Palmer and Gough (P&G)

Palmer and Gough (2007) examined the difference in the attribution of importance of defective education as an explanation for criminal behavior

between three types of “offenders”:

 $(n_i = 20, 20, 31)$

ANOVA Example: Comparisons of 3 Means

Example Palmer and Gough (P&G)

Palmer and Gough (2007) examined the difference in the attribution of importance of defective education as an explanation for criminal behavior between three types of “offenders”:
(1) non-offenders, (2) property offenders, and (3) person offenders.

ANOVA

$$y_j = \mu_1 d_{1j} + \mu_2 d_{2j} + \mu_3 d_{3j} + \epsilon_j,$$

where μ_i ($i = 1, 2, 3$) is the mean in group i ,
 d_{ij} is a dummy variable denoting group membership,
 ϵ_j is an error term and $\epsilon_j \sim N(0, \sigma^2)$.

$(n_i = 20, 20, 31)$

Hypotheses of interest

Example Palmer and Gough (P&G)

Palmer and Gough (2007) expect that

- non-offenders (1) attribute more importance to defective education for explaining crime than the other two offenders (2 & 3):
i.e., $\mu_1 > \mu_2$ and $\mu_1 > \mu_3$,
- property offenders (2) attribute more importance to defective education than person offenders (3):
i.e., $\mu_2 > \mu_3$.

$$H_1: \mu_1 > \mu_2 > \mu_3.$$

Conclusions w.r.t. hypothesis of interest $H_1 : \mu_1 > \mu_2 > \mu_3$

Example 3 means

With post-hoc test (e.g., Bonferroni) and on basis of the sample means, we could say a bit more.

But, what if

- $\mu_1 \neq \mu_2, \mu_1 \neq \mu_3, \mu_2 = \mu_3,$
- $\bar{y}_1 > \bar{y}_2,$

which implies $\mu_1 > \mu_2 = \mu_3$.

But also not interested in this.

Furthermore, inconsistent results are possible (like in P&G example):

$$\mu_1 = \mu_2, \mu_2 = \mu_3, \text{ and } \mu_1 \neq \mu_3,$$

which is logically impossible, since the first two imply $\mu_1 = \mu_3$.

Using planned contrasts would overcome this, but still = restrictions then.

NHST

Example Palmer and Gough

```
PandG_data <- read.table("Data_PalmerAndGough.txt",
                          header=TRUE)
PandG_data$group <- factor(PandG_data$group)
pairwise.t.test(PandG_data$Importance, PandG_data$group,
                p.adj = 'bonferroni')

##
## Pairwise comparisons using t tests with pooled SD
##
## data: PandG_data$Importance and PandG_data$group
##
##      1      2
## 2 0.191 -
## 3 0.011 1.000
##
## P value adjustment method: bonferroni
```


Exploration and Confirmation

Exploration (like in post hoc tests and possibly AIC)

Evaluate all possible pairs/subsets of means whether significant different (" $\neq$ ") or not (" $=$ ").

For example, when $k = 3$:

$$H_{0E} : \mu_1 = \mu_2 = \mu_3$$

$$H_{1E} : \mu_1 = \mu_2, \mu_3$$

$$H_{2E} : \mu_1, \mu_2 = \mu_3$$

$$H_{3E} : \mu_1 = \mu_3, \mu_2$$

$$H_a : \mu_1, \mu_2, \mu_3$$

When $k = 5$, there are even 52 hypotheses.

Confirmatory methods

Most researchers are able to specify “order-restricted” / “informative” / “theory-based” hypotheses, like $H_1 : \mu_1 > \mu_2 > \mu_3$.
Use prior knowledge and/or expertise in hypothesis.

Note: 'model' refers to hypothesis.

GORIC

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

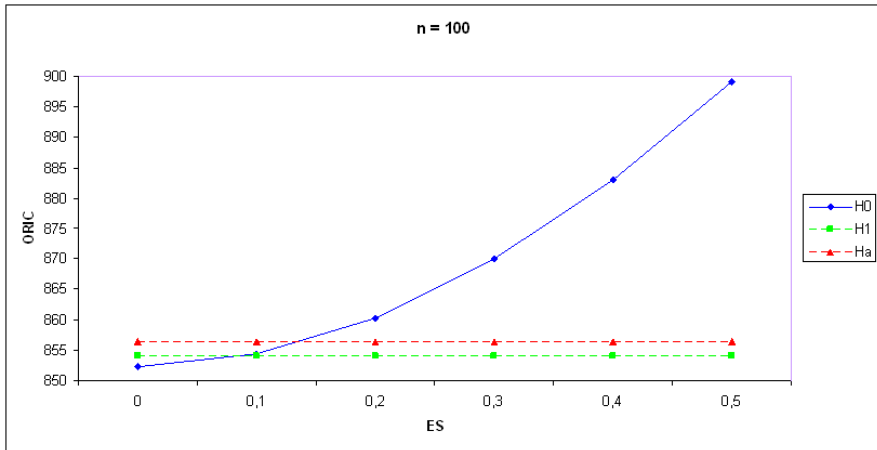
$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC – will be explained later

Model	Fit	Complexity	GORIC	GORIC weights
H_0	-196.36	2.00	396.71	0.02
H_1	-191.89	2.81	389.41	0.75
H_u	-191.89	4.00	391.79	0.23

Confirmatory methods (e.g., GORIC) have more “power” than their exploratory counterparts (e.g., AIC; cf. one- vs two-sided testing).

Confirmation more power: 1 data set. GORIC values for 3 groups, effect size ES , and $n = 100$ observations per group



ANOVA Example: Comparisons of 3 Means

Simple, hypothetical example

Examine the difference in happiness
between three types of “treatments”:
(A) treatment A, (B) treatment B, and (C) no treatment / control.

DIY: Think of one or more theory-based hypotheses.
(in terms of μ_A , μ_B , and/or μ_C .)

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GORICA

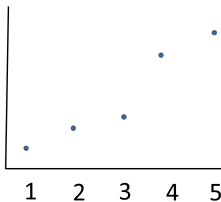
GORIC(A) in JASP

End & Extra

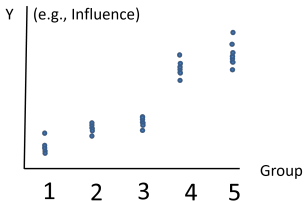
Intermezzo: Balance Fit and Complexity (1/6)

Example 5 means

Data for 5 groups – Oversimplified representation

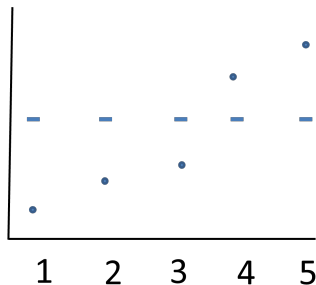


Data for 5 groups – Better representation (scatter of data points)



Intermezzo: Balance Fit and Complexity (2/6)

Example 5 means

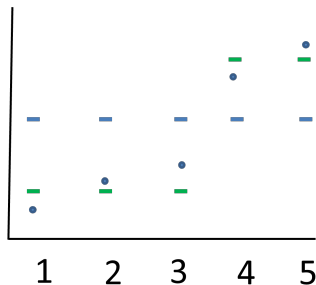


$$\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5 = \mu$$

parameters: 1 mean parameter (i.e., μ)
Fit: Bad fit

Intermezzo: Balance Fit and Complexity (3/6)

Example 5 means



$$\mu_1 = \mu_2 = \mu_3 = \mu_{\text{low}}$$

$$\mu_4 = \mu_5 = \mu_{\text{high}}$$

parameters:

2 mean parameters (i.e., μ_{low} and μ_{high})

Fit:

Better than with one parameter, even good fit.

Intermezzo: Balance Fit and Complexity (6a/6)

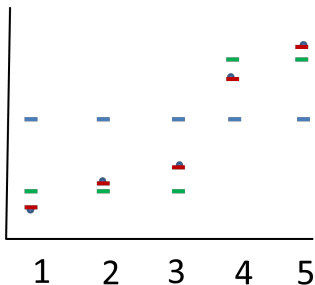
Order-restrictions

Previous slides demonstrated balancing fit and complexity for equality ('=') restrictions.

For inequality/order ('<', '>', and possible some '=') restrictions, the calculation of the fit and complexity is more complex :-), but the idea of balancing is the same.

Intermezzo: Balance Fit and Complexity (6b/6)

Order-restrictions: From “Example 5 means” to “Example 2 means”

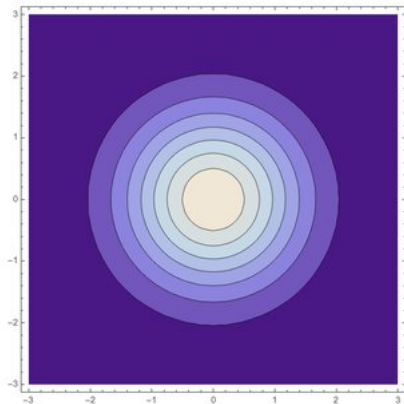
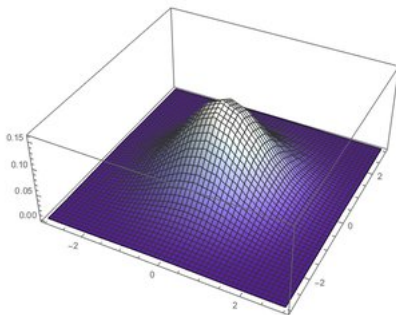


In theory-based hypotheses, we also incorporate **order-restrictions** (e.g., $\mu_1 < \mu_2$). Then, helpful to look at likelihood using **contour plots** (not scatter plot, as done here).

For ease, I will next use **2 means** (instead of 5).

Intermezzo: Contour plot (1/2)

Example 2 means

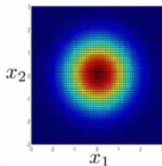
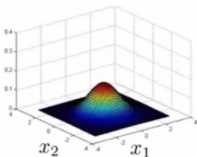


Intermezzo: Contour plot (2/2)

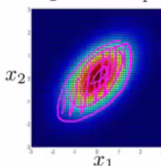
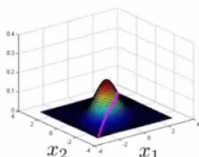
Example 2 means

Multivariate Gaussian (Normal) examples

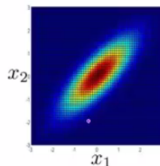
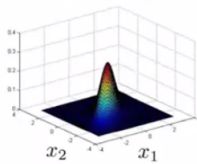
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$$



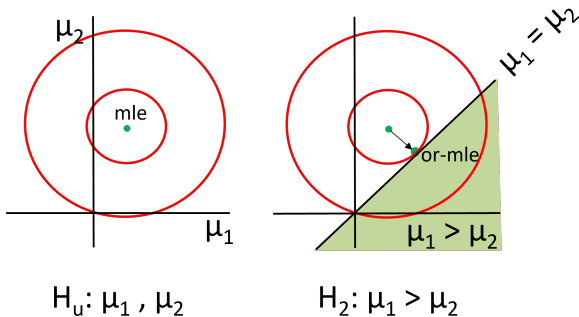
$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$$



Andreas

Idea fit: order-restricted maximum likelihood (or-ml)

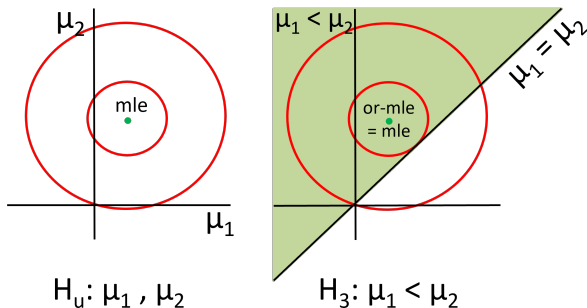
Example 2 means (with or-mle $\neq$ mle)



mle not in H_2 : Find highest likelihood in allowable (= green) space.
The resulting estimated mean $\hat{\mu} = (\hat{\mu}_1, \hat{\mu}_2)$ is referred to as or-mle.

Note: if in agreement, or-mle = mle

Example 2 means with or-mle = mle



mle in H_3 : highest likelihood is in allowable (= green) space.
Hence, or-mle = mle.

Idea fit

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_1 > \mu_2 < \mu_3,$$

$$H_3 : \mu_1 < \mu_2 < \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

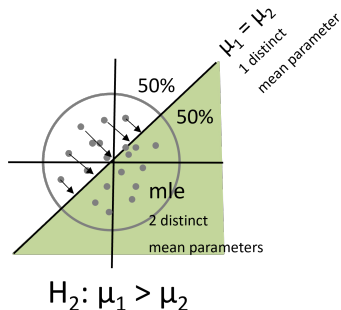
```
## group1 group2 group3
## 11.95  9.75  8.77
```

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_0	-196.36	2.00	396.71	0.01
H_1	-191.89	2.81	389.41	0.56
H_2	-192.34	3.19	391.05	0.25
H_3	-196.36	2.81	398.34	0.01
H_u	-191.89	4.00	391.79	0.17

Idea complexity: penalty (PT)

Example 2 means: Contour distribution



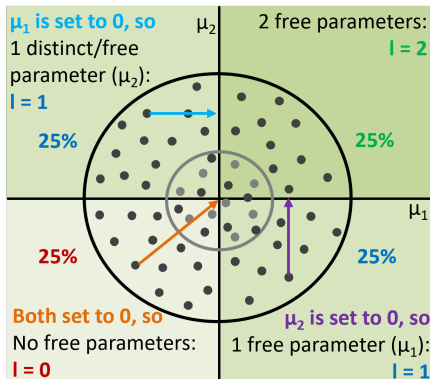
Sampling from distribution with $\mu_1 = \mu_2 = 0$. (Note: not likelihood!)

Contour denotes distribution for a variance value.

Choice variance does not matter, see next slide.

Extra: Levels and level probabilities

Example 2 means



$$H_4: \mu_1 > 0, \mu_2 > 0$$

Level (l): number of distinct/free parameters.

Level probability (LP_l): probability that there are l levels.

$$PT_4 = 1 + 0.25 \times 0 + (0.25 + 0.25) \times 1 + 0.25 \times 2 = 2.$$

Idea complexity

Example Palmer and Gough, that is, 3 means

$$H_0 : \mu_1 = \mu_2 = \mu_3,$$

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_1 > \mu_2 < \mu_3,$$

$$H_3 : \mu_1 < \mu_2 < \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

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GORIC: Lowest value is best

GORIC is like AIC expected distance from the truth (KL-distance).
One wants smallest distance.

Or: One wants the smallest 'misfit + complexity'.

Note: $\text{GORIC} = -2 \text{ fit} + 2 \text{ complexity} = 2 \text{ misfit} + 2 \text{ complexity}$

Hence, smallest value is best.

GORIC

Example Palmer and Gough

- $H_0 :$ $\mu_1 = \mu_2 = \mu_3$ (include only when of interest)
 $H_1 :$ $\mu_1 > \mu_2 > \mu_3$
 $H_2 :$ $\mu_2 > \mu_1 > \mu_3$ (if competing theory/hypothesis)
 $H_u :$ μ_1, μ_2, μ_3 .

GORIC

Model	Fit	Complexity	GORIC
H_0	-196.36	2.00	396.71
H_1	-191.89	2.81	389.41
H_2	-193.70	2.81	393.03
H_u	-191.89	4.00	391.79

Hands-on/Demo (1a): GORIC

- Go to <https://github.com/rebeccakuiper/Tutorials>:
 - Click on green button called Code.
 - Download zip (last option in list).
 - Unzip it on your machine (that folder is now your working dir.).
- Start Rstudio. Optional: make project.
- Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_1_GORIC_Unc_ANOVA_restriktor.R**' ('Hands-on files').
- Install packages and load them.
- Read and inspect data.
Use **Data_PalmerAndGough.txt** and/or Data_Lucas.txt.
- Run model (lm()).
- Specify hypotheses (make up your own).
Note: Use names used in the model.
- Run goric().
- Inspect and interpret output.
Note: 'GORIC weights' will be explained next.



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End & Extra

Interpretation: GORIC weights

GORIC values

GORIC values cannot be interpreted, only compared:
Smallest is best.

GORIC weights (w_m)

w_m quantifies relative support for H_m versus others in the set.
Values between 0 and 1, and they sum to 1.
The bigger, the better.

Illustration of the GORIC weights (w_m)

Example Palmer and Gough

$H_0 : \mu_1 = \mu_2 = \mu_3$ (include only when of interest)

$$H_1 : \mu_1 > \mu_2 > \mu_3$$
$$H_2: \mu_2 > \mu_1 > \mu_3 \text{ (if competing theory/hypothesis)}$$
$$H_u : \quad \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

H_1 is $0.68 / 0.11 \approx 6.11$ times more supported than H_2 .

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Read and inspect data.
- Run model (`lm()`).
- Specify hypotheses (make up your own).
Note: Use names used in the model.
- Run `goric()`.
- Inspect and interpret output: Focus on GORIC weights.

GORIC output and interpretation

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.68
H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

H_1 and H_2 are not weak (nl, better than H_u - will be explained in a bit).

H_1 is $0.68 / 0.11 \approx 6.11$ times more supported than H_2 .

(Two informative) hypotheses of interest

$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$ (include only when of interest)

$H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$

$H_2 : \mu_3 > \mu_1 > \mu_4 = \mu_5 > \mu_2$

$H_u : \mu_1, \mu_2, \mu_3, \mu_4, \mu_5$ (included as failsafe)

Descriptive statistics of Lucas' Data

Group	Mean Influence	SD	n
1 (male, random)	2.33	1.86	30
2 (female, random)	1.33	1.15	30
3 (male, selected)	3.20	1.79	30
4 (female, selected)	2.23	1.45	30
5 (female, selected+)	3.23	1.50	30

GORIC weights (w_m)

Lucas

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-278.05	3.46	563.02	0.90
H_2	-281.76	3.14	569.79	0.03
H_u	-278.05	6.00	568.10	0.07

H_1 is $0.9 / 0.03 \approx 29.59$ times more supported than H_2 .
Thus, there is quite some evidence that H_1 is the best of this set.

Note:

H_1 is $0.9 / 0.07 > 1$ times more supported than H_u .
Hence, H_1 not weak and thus comparable to the other
(weak and non-weak) informative hypotheses (as explained in a bit).

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End & Extra

Include “unconstrained” hypothesis

If set of hypotheses does not contain a reasonable/good one:
 Select the best of set of weak hypotheses.
 E.g.: $w_1 = .8$ and $w_2 = .2$.

Prevent choosing a weak hypothesis

Include unconstrained hypothesis H_u (or H_a):

$$H_1 : \quad \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \quad \mu_2 > \mu_1 > \mu_3$$

$$H_u : \quad \mu_1, \mu_2, \mu_3,$$

(i.e., *no restrictions*).

H_u highest fit but also most complex, thus failsafe/safeguard.

E.g.: $w_1 = .08$, $w_2 = .02$, and $w_u = .90$.

Use of H_u

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

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H_2	-193.70	2.81	393.03	0.11
H_u	-191.89	4.00	391.79	0.21

If at least one informative hypothesis not weak (i.e., $w_m > w_u$ or $w_m/w_u > 1$), then compare informative hypotheses.

Hence: H_u is only a failsafe, not another hypothesis of interest.

Use of H_u

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

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Use of H_u

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_2 : \mu_2 > \mu_1 > \mu_3$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights	without Hu
H_1	-191.89	2.81	389.41	0.68	0.86
H_2	-193.70	2.81	393.03	0.11	0.14
H_u	-191.89	4.00	391.79	0.21	NA

H_1 is not weak and is $0.68 / 0.11 = 0.86 / 0.14 \approx 6.11$ times more supported than competing hypothesis H_2 .

H_1 vs H_u

Palmer & Gough

What if only one informative hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

GORIC

Model	Fit	Complexity	GORIC	GORIC weights
H_1	-191.89	2.81	389.41	0.77
H_u	-191.89	4.00	391.79	0.23

H_1 is $0.77 / 0.23 \approx 3.27$ times more supported than H_u .

BUT: H_u includes H_1 .

So, support for H_u contains support for H_1 .

H_1 vs H_u

Palmer & Gough

What if only one informative hypothesis:

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_u : \mu_1, \mu_2, \mu_3.$$

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H_1 vs H_u

Palmer & Gough

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BUT: H_u includes H_1 .

So, support for H_u contains support for H_1 .

Alternative failsafe: Complement of H_m

Alternatively (in the case of one hypothesis of interest)

Evaluate hypothesis of interest against its complement;
that is, all other possible hypotheses.

More powerful than against the unconstrained if H_m has maximum fit.

Reference:

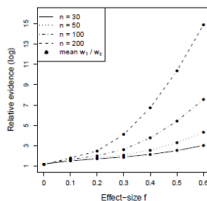
Vanbrabant, L., Van Loey, N., and Kuiper, R. M. (2020). Evaluating a Theory-Based Hypothesis Against Its Complement Using an AIC-Type Information Criterion With an Application to Facial Burn Injury. *Psychological Methods*, 25(2), 129-142.

<https://doi.org/10.1037/met0000238>

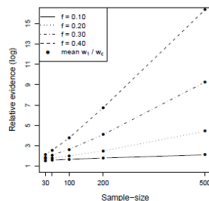
Alternative failsafe: Complement of H_m

Based on simulation

vs complement

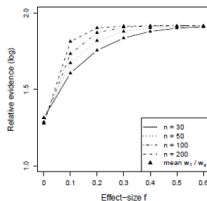


(a)

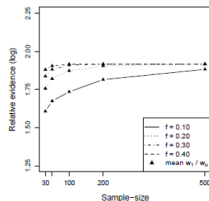


(b)

vs unconstrained



(c)



(d)

R code: complement

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORIC

```

H1 <- 'group1 > group2 > group3'
# vs its complement (default in case of one hypothesis)
#
# GORIC (using goric function in restriktor package)
library(restriktor)
set.seed(123) # Set seed value
goric.PandG_1c <- goric(fit.PandG,
                        hypotheses = list(H1))
    
```


Extra: complement & 'Heq' (3/4)

Palmer & Gough

R output (partly)

```
## restriktor (0.6-50): generalized order-restricted information criterion:
##
## Results:
##      model      loglik  penalty    goric  loglik.weights  penalty.weights
## 1      Heq   -193.703    3.000  393.406         0.123         0.452
## 2      H1   -191.893    3.500  390.786         0.753         0.274
## 3 complement -193.703    3.500  394.406         0.123         0.274
##      goric.weights  goric.weights_without_heq
## 1              0.188
## 2              0.698
## 3              0.114
##
## Conclusion:
## - The order-restricted hypothesis H1 is the best in the set, as it has the h
## * 'H1' is 6.109 times more supported than 'complement'.
```


Hands-on/Demo (2): GORIC - H_m vs complement

Let's practice.

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Optional: Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_2_GORIC_Compl_ANOVA_restriktor.R**'.
- Read and inspect data.
Use Data_PalmerAndGough.txt and/or **Data_Lucas.txt**.
- Run model (lm()).
- Specify hypotheses (make up your own). E.g., for Lucas:

$$H_1 : \quad \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$

Note: Use names used in the model.

- Run goric()
- Inspect and interpret output.

GORICA

GORIC: Normal linear models

GORIC can easily be applied to normal linear models (e.g., ANOVA models or regression models).

GORICA

GORIC: Normal linear models

GORIC can easily be applied to normal linear models (e.g., ANOVA models or regression models).

GORIC: Other statistical models

In case of other statistical models (e.g., a SEM model), more cumbersome to calculate maximized order-restricted log likelihood and thus GORIC.

GORICA: All statistical models

Therefore, GORICA: asymptotic expression for GORIC.
Can be used for all types of statistical models.

Reference:

Altınışık, Y., Van Lissa, C. J., Hoijtink, H., Oldehinkel, A. J., and Kuiper, R. M. (2021). Evaluation of inequality constrained hypotheses using a generalization of the AIC. *Psychological Methods*, 26(5), 599–621.

<https://doi.org/10.1037/met0000406>

GORICA

Similarities with GORIC

- Form: $GORICA_m = -2 \text{ fit} + 2 \text{ complexity}$.
- Broad type of restrictions.

Differences compared to GORIC

- Uses asymptotic expression of the likelihood (is a normal):
can therefore be easily applied to all types of statistical models.
Disadvantage: might work less well in case of small samples.
- Does not need data set; mle's and their covariance matrix suffice.
- Can leave out nuisance parameters (i.e., not part of hypotheses).

Note

In case of normal linear models or, otherwise, for not too small samples:
GORICA weights = GORIC weights.

GORICA

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In case of normal linear models or, otherwise, for not too small samples:
GORICA weights = GORIC weights.

R code: GORICA

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORICA: type = "gorica" (not default for lm objects)

```

H1 <- 'group1 > group2 > group3'
# vs its complement (default in case of one hypothesis)
#
# GORICA (using goric function in restriktor package)
library(restriktor)
set.seed(123) # Set seed value
gorica.PandG_C <- goric(fit.PandG,
                        hypotheses = list(H1),
                        type = "gorica") # needed if lm object
    
```

GORICA

Palmer & Gough

$$H_1 : \mu_1 > \mu_2 > \mu_3,$$

$$H_c : \text{not } H_1.$$

GORIC

Model	Fit	Complexity	GORICA	GORICA weights
H1	-1.90	1.81	7.42	0.79
complement	-2.34	2.69	10.06	0.21

H_1 is $0.79 / 0.21 \approx 3.74$ times more supported than its complement, that is, any other hypothesis.

Note: GORIC(A) weights are the same (ratio may differ a bit).

GORICA

Palmer & Gough

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GORIC

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Note: GORIC(A) weights are the same (ratio may differ a bit).

Hands-on/Demo (3): GORICA

Let's practice.

- If needed: Start Rstudio again (optional: make project) and then also load packages again.
- Optional: Open 'Tutorial_GORIC_restriktor_ANOVA.html', 'PalmerAndGough_and_Lucas.R', and/or '**Hands-on_3_GORICA_UncAndCompl_ANOVA_restriktor.R**'.
- Read and inspect data.
Use Data_PalmerAndGough.txt and/or **Data_Lucas.txt**.
- Run model (lm()).
- Specify hypotheses (make up your own). E.g., for Lucas:

$$H_1 : \mu_5 = \mu_3 > \{\mu_1, \mu_4\} > \mu_2$$

Note: Use names used in the model (or overwrite those).

- Run goric(); now, add (since lm object):
`type = "gorica"`
- Inspect and interpret output.

Power improvement

Power improvement via

- Using order restrictions (cf. one-sided vs two-sided testing).
- Having limited set of hypotheses.
Note: Null hypothesis can be one of them, but only if of interest!
Try to think outside of the null-hypothesis box :-).
- (if:) Specifying restrictions for multiple outcomes within each informative hypothesis.

Power improvement

Power improvement via

- Using order restrictions
(cf. one-sided vs two-sided testing).
- Having limited set of hypotheses.
Note: Null hypothesis can be one of them, but only if of interest!
Try to think outside of the null-hypothesis box :-).
- (if:) Specifying restrictions for multiple outcomes within each informative hypothesis.

If you do not have a priori expectations and sample size is large enough:

- 1) You could use part of your data to generate one or more hypotheses.
- 2) This/These can then be evaluated in the other part of the data.

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Generalized Order-Restricted Information Criterion (GORIC)

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GORICA

GORIC(A) in JASP

End & Extra

GORIC in JASP: screenshot

Palmer & Gough

Descriptives

T-Tests

ANOVA

Mixed Models

Regression

Frequencies

Factor

Bain

SEM

R (Beta)

Order Restricted Hypotheses

Enter each restriction of one hypothesis on a new line, e.g.,
factorLow == factorMid
factorMid < factorHigh
where 'factor' is the factor (or covariate) name and 'Low'/'Mid'/'High' are the factor level names.
Click on the 'plus' icon to add more hypotheses.
Click the information icon for more examples.

Syntax settings

☐ Include intercept
☐ Show available coefficients

Set for all models

☐ Model summary
☐ Marginal means
☐ Informed hypothesis tests

Model 1

+

group1 > group2 > group3

Ctrl + Enter to apply. Click on the blue button above for help on the restriction syntax

☐ Summary for Model 1
☐ Marginal means for Model 1
☐ Informed hypothesis tests for Model 1

Results

ANOVA

ANOVA - Importance

Cases	Sum of Squares	df	Mean Square	F	p
group	123.960	2	61.980	4.554	0.014
Residuals	925.418	68	13.609		

Note. Type III Sum of Squares

Order Restricted Hypotheses

Model Comparison

Model Comparison Table

Model	Log-likelihood	Penalty	GORIC	Weight	Weights ratio
Model 1	-191.893	2.814	389.415	0.789	3.731
Complement	-192.338	3.686	392.048	0.211	1.000

Note. Weights ratios indicate the relative weight for each model against the "complement" model. GORIC = Generalized Order-Restricted Information Criterion (Kuiper, Hoijtink, & Silvapulle, 2011).

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GORIC and GORICA in JASP

GORIC or GORICA in the following statistical models (as part of that model)

The image displays three screenshots of the JASP statistical software interface, illustrating where GORIC and GORICA can be applied within different models.

- Top Screenshot (ANOVA):** The left sidebar shows the 'Order Restricted Hypotheses' option highlighted with a red circle. The main window displays the 'ANOVA' model results table.
- Middle Screenshot (Repeated Measures ANOVA):** The left sidebar shows the 'Order Restricted Hypotheses' option highlighted with a red circle. The main window displays the 'Repeated Measures ANOVA' model results table.
- Bottom Screenshot (ANCOVA):** The top dropdown menu is set to 'ANCOVA', which is highlighted with a red circle. The left sidebar shows the 'Order Restricted Hypotheses' option highlighted with a red circle. The main window displays the 'ANCOVA' model results table.

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GORICA

GORIC(A) in JASP

End & Extra

If you have your own data

- What is your research question?
- What is your theory / expectation?
- What is your statistical hypothesis?
- Is there a competing statistical hypothesis?

- Are you able to specify your statistical hypothesis/-es?
- How will you evaluate it/them?

The End

GORIC(A)

Thanks for listening!

Are there any questions?

Websites

<https://github.com/rebeccakuiper/Tutorials>

www.uu.nl/staff/RMKuiper/Extra1

www.uu.nl/staff/RMKuiper/Extra2

informative-hypotheses.sites.uu.nl/software/goric/

E-mail

r.m.kuiper@uu.nl

Notes

Note on type of hypotheses: Magnitude-hypotheses

Hypotheses can regard height of effect size.

In the case of Cohen's d , one could evaluate:

$$H_1 : d < 0,$$

$$H_2 : d > 0,$$

$$H_3 : d > 0.2,$$

$$H_4 : d > 0.5,$$

$$H_5 : d > 0.8.$$

$$H_1 : d < 0,$$

$$H_2 : 0 < d < 0.2,$$

$$H_3 : 0.2 < d < 0.5,$$

$$H_4 : 0.5 < d < 0.8,$$

$$H_5 : d > 0.8.$$

Now, overlapping hypotheses.

Now, range restrictions
(complexity as if equalities).

Or better: One of these versus its complement.

Or incorporate such cut-off values when inspecting ordering of parameters (say, means).

Note on possibilities multiple studies

- Update GORIC(A) values & weights.

More data collected: (re-)calculate.

- Update hypotheses.

If no expectations (or no support), use exploration and confirmation:

First data set (or a part of it) generates one or more hypotheses.

Other data set (or part) used to determine evidence / support.

Download 'Tutorial_GORIC_restriktor_UpdateHypo.html' and/or

'Hands-on_4_GORIC_UpdateHypo_restriktor.R' from

<https://github.com/rebeccakuiper/Tutorials>.

- Aggregate evidence for hypotheses.

Aggregate the support for theories (diverse designs allowed).

Bear in mind: Meta-analysis aggregates parameter estimates or effect sizes which need to be comparable (often same designs required).

Download 'Tutorial_GORIC_restriktor_evSyn.html' and/or

'Hands-on_4_GORIC_evSyn_restriktor.R' from

<https://github.com/rebeccakuiper/Tutorials>.

Extra material (1/2)

- GORICA on logistic Regression Model
 - Article: doi.org/10.1037/met0000406
- GORICA on SEM
 - Article: www.tandfonline.com/doi/full/10.1080/10705511.2020.1836967.
 - R scripts: github.com/rebeccakuiper/GORICA_in_SEM.
- GORICA on cross-lagged panel model (CLPM)
 - Article: doi.org/10.1111/bjep.12455.
 - R scripts: github.com/rebeccakuiper/GORICA_in_CLPM.
- GORICA on Random-Intercept CLPM (RI-CLPM)
 - Article: Sukpan, C. and Kuiper, R.M. (2023). How to evaluate causal dominance hypotheses in lagged effects models.
 - R scripts: github.com/rebeccakuiper/GORICA%20in%20RI-CLPM.
- GORICA for EffectLiteR
 - R scripts: <https://github.com/rebeccakuiper/Tutorials/tree/main/GORICA%20for%20EffectLiteR>.

Note: On github site, go to Code (green button) and download zip.

Note on conditional error probability (1/3)

using PMPs and GORIC(A) weights

H_m	weights
H_1 : Sex Match	.04
H_2 : Gender Role Match	.81
H_3 : Sex Mismatch	.01
H_4 : Gender Role Mismatch	.00
H_u :	.14

The conditional error probability for preferred hypotheses H_2 is $1 - .81 = .19$.

Note on conditional error probability (3/3)

using PMPS and GORIC(A) weights

What does conditional error probability reflect when hypotheses overlap? Notably, when hypotheses overlap they can also share/divide support (like H_u does with any hypothesis).

My advise:

Only use error probabilities if one hypothesis versus its complement.
(Or when you are sure that there is no overlap in hypotheses;
and they preferably do cover all possibilities)

H_m	weights
H_2 : Gender Role Match	.82
H_c : complement H_2	.18

Population Values in Simulated Data Sets when $k = 3$

Values of Population Means (μ_i) and Standard Deviations (σ_i) Used to Simulate 1000 Data Sets

$k = 3$	Case Number ('ES')	μ_1	μ_2	μ_3	ES (Cohen)
	0	0.00	0.00	0.00	
	1	-0.10	0.00	0.10	.1
	2	-0.20	0.00	0.20	.2
	3	-0.30	0.00	0.30	.3
	4	-0.40	0.00	0.40	.4
	5	-0.50	0.00	0.50	.5
		σ_1	σ_2	σ_3	
	0 - 5	1.00	1.00	1.00	

NB in Case 0 $H_{30C} : \mu_1 = \mu_2 = \mu_3$ is true
in Cases 1 to 5 $H_{31C} : \mu_1 < \mu_2 < \mu_3$ is true.

Results Confirmation ($k = 3$ and $n = 50$)

$$H_{30C} : \mu_1 = \mu_2 = \mu_3$$

$$H_{31C} : \mu_1 < \mu_2 < \mu_3$$

$$H_{32C} : \mu_1 = \mu_2 < \mu_3$$

$$H_{33C} : \mu_1 < \mu_2 > \mu_3$$

$$H_{34C} : \mu_1, \mu_2, \mu_3$$

1 – Type I error (here):

preferring the true null hypothesis,
given the set of hypotheses (!).

'ES'	Method	'Prior'	H_{30C}	H_{31C}	H_{32C}	H_{33C}	H_{34C}
0	ORIC		0.662	0.065	0.119	0.120	0.034
0	BMS	3	0.865	0.012	0.075	0.036	0.012
0	BMS	2	0.774	0.022	0.124	0.058	0.022
0	BMS	1	0.656	0.047	0.147	0.120	0.033
2	ORIC		0.148	0.506	0.311	0.029	0.006
2	BMS	3	0.299	0.299	0.371	0.031	0.001
2	BMS	2	0.252	0.328	0.381	0.040	0.000
2	BMS	1	0.125	0.494	0.321	0.063	0.000
5	ORIC		0.000	0.955	0.044	0.001	0.000
5	BMS	3	0.000	0.887	0.112	0.001	0.000
5	BMS	2	0.001	0.890	0.107	0.002	0.000
5	BMS	1	0.000	0.928	0.070	0.002	0.000

Results Confirmation ($k = 3$ and $n = 50$)

$$H_{30C} : \mu_1 = \mu_2 = \mu_3$$

$$H_{31C} : \mu_1 < \mu_2 < \mu_3$$

$$H_{32C} : \mu_1 = \mu_2 < \mu_3$$

$$H_{33C} : \mu_1 < \mu_2 > \mu_3$$

$$H_{34C} : \mu_1, \mu_2, \mu_3$$

1 – Type II error (here):
preferring the true hypothesis,
given the set of hypotheses (!).

'ES'	Method	'Prior'	H_{30C}	H_{31C}	H_{32C}	H_{33C}	H_{34C}
0	ORIC		0.662	0.065	0.119	0.120	0.034
0	BMS	3	0.865	0.012	0.075	0.036	0.012
0	BMS	2	0.774	0.022	0.124	0.058	0.022
0	BMS	1	0.656	0.047	0.147	0.120	0.033
2	ORIC		0.148	0.506	0.311	0.029	0.006
2	BMS	3	0.299	0.299	0.371	0.031	0.001
2	BMS	2	0.252	0.328	0.381	0.040	0.000
2	BMS	1	0.125	0.494	0.321	0.063	0.000
5	ORIC		0.000	0.955	0.044	0.001	0.000
5	BMS	3	0.000	0.887	0.112	0.001	0.000
5	BMS	2	0.001	0.890	0.107	0.002	0.000
5	BMS	1	0.000	0.928	0.070	0.002	0.000

Extra: Prior choice for BMS

Prior

$$P(\boldsymbol{\mu}, \sigma^2) = P(\mu_1) \times \dots \times P(\mu_k) \times P(\sigma^2),$$

where $P(\mu_i) = P(\mu) \forall i = 1, \dots, k$.

$P(\mu)$ is a data-based normal distribution, with parameters β_0 and τ_0^2 , and $P(\sigma^2)$ an inverse chi-squared distribution (see Klugkist et al., 2005)

β_0 and τ_0^2 data-based hyperparameters

For each μ_i ($i = 1, \dots, k$) a credibility interval is computed:

estimated group mean $\pm$ $PV \times$ *the group standard error*.

The credibility interval for μ_i is represented by " $[LB_i, UB_i]$ ".

Lowest lower bound: LB_{min} .

Highest upper bound: UB_{max} .

Then, $\beta_0 = \frac{LB_{min} + UB_{max}}{2}$ and $\tau_0^2 = \left[\frac{UB_{max} - LB_{min}}{2} \right]^2$.

Three types of prior vagueness: BMS PV 1, BMS PV 2 and BMS PV 3